

YSU CS2020 — Machine Learning

Assignment 3: Decision Trees & Ensemble Methods

Wine Quality Prediction Using CART Decision Trees and Random Forests

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Course: YSU CS2020 — Machine Learning

Type: Self-Study Project | Duration: 3 Weeks

Dataset: UCI Wine Quality (Red Wine, 1,599 samples)

Abstract

This report presents a comprehensive study of Decision Trees and Random Forests, two fundamental machine learning algorithms for classification tasks. The main objective of this work is to implement both methods from scratch, analyse their behaviour on a real-world dataset, and compare their performance with standard implementations from the sklearn library.

The study is conducted on the UCI Red Wine Quality dataset, which contains physicochemical properties of wine samples along with quality ratings. Three main experiments were performed: (1) model comparison between custom and sklearn implementations, (2) hyperparameter tuning to analyse model behaviour and overfitting, and (3) feature importance analysis to understand which variables contribute most to predictions.

The results show that the Random Forest model significantly outperforms a single Decision Tree in terms of generalisation performance. While the Decision Tree achieves a test accuracy of approximately 57%, the Random Forest reaches around 63-64%, demonstrating the effectiveness of ensemble methods in reducing overfitting. Additionally, the analysis reveals that only a subset of features is sufficient to achieve near-optimal performance.

Overall, this study highlights the importance of ensemble learning, the bias-variance tradeoff, and the practical usefulness of feature importance analysis in real-world machine learning problems.

1. Introduction

Decision Trees are one of the most intuitive and interpretable machine learning models. They work by recursively splitting the data into smaller subsets based on feature values, forming a tree-like structure. Each internal node represents a decision rule, and each leaf node corresponds to a class prediction. Because of their structure, decision trees are easy to

visualise and understand, which makes them particularly useful in applications where interpretability is important.

However, despite their advantages, decision trees suffer from a major drawback: they tend to overfit the training data. This happens because the tree keeps splitting the data until it perfectly fits the training set, capturing noise instead of general patterns. As a result, performance on unseen data (test set) often degrades significantly.

Random Forests were introduced as a solution to this problem. Instead of relying on a single tree, Random Forest builds multiple decision trees using random subsets of data and features, and combines their predictions. This ensemble approach reduces variance and leads to better generalisation.

The motivation for this study is to understand:

- How much Random Forest improves over a single Decision Tree
- How hyperparameters affect model performance
- Which features are most important in predicting wine quality

The research questions are:

1. How does Random Forest compare to a single Decision Tree in terms of accuracy and overfitting?
2. What are the optimal hyperparameters for this dataset, and when does overfitting occur?
3. Which features contribute most to prediction, and how many features are actually needed?

The dataset used is the UCI Red Wine Quality dataset, which contains 1,599 samples with 11 numerical features describing chemical properties such as acidity, sugar content, and alcohol level. This dataset was chosen because it represents a real-world classification problem with meaningful features that can be interpreted.

The main contribution of this work is a complete implementation of Decision Trees and Random Forests from scratch, along with a detailed experimental analysis connecting theory to practice.

All experiments were conducted with an 80/20 train-test split.

2. Theoretical Background

2.1 Decision Trees (CART)

The CART (Classification and Regression Trees) algorithm constructs a binary tree by recursively splitting the dataset into two parts at each node. The goal is to find the split that best separates the classes.

At each step, the algorithm evaluates all possible splits across all features and selects the one that maximises **information gain**, which measures how much uncertainty is reduced after the split.

Two common impurity measures are used:

Gini Impurity: This measures how often a randomly chosen element would be incorrectly classified.

$$Gini(D) = 1 - \sum p_k^2$$

Entropy: This measures the uncertainty in the dataset.

$$H(D) = - \sum p_k \log_2(p_k)$$

The **Information Gain** is defined as:

$$IG = H(parent) - \sum \frac{n_i}{n} H(child_i)$$

The algorithm selects the split that maximises this value.

The recursion stops when:

- All samples belong to the same class
- Maximum depth is reached
- Minimum number of samples is not satisfied
- No split improves information gain

Advantages:

- Easy to interpret
- Works with numerical data
- No need for scaling

Disadvantages:

- Highly sensitive to noise
- Prone to overfitting
- High variance

2.2 Random Forests

A Random Forest is an ensemble of decision trees trained using two sources of randomness to decorrelate them:

- Bagging (Bootstrap Aggregating): Each tree is trained on a bootstrap sample — a random sample with replacement of the training data of the same size as the original. Each bootstrap sample includes on average ~63.2% of unique training points.
- Feature randomness: At each split, instead of considering all features, only a random subset of size $\sqrt{n_features}$ is considered. This further decorrelates the trees and prevents any single dominant feature from appearing in every tree.

Predictions are made by majority vote across all trees. Because averaging reduces variance while leaving bias largely unchanged, Random Forests typically generalise much better than individual deep trees. The bias-variance decomposition explains this: a single deep tree has low bias but high variance; the forest preserves the low bias while dramatically reducing variance through averaging.

Random Forests are preferred over single trees when predictive performance matters more than interpretability, when the dataset has many features, and when there is sufficient data to train many trees.

2.3 Information Theory

Entropy, borrowed from Shannon's information theory, measures the uncertainty of a probability distribution. A node with equal class proportions has maximum entropy; a pure node has entropy zero. Maximising information gain is equivalent to minimising the weighted average entropy of the child nodes, which corresponds to choosing splits that produce the purest possible partitions. Gini impurity has a similar geometric interpretation but avoids the logarithm computation, making it slightly faster in practice — both criteria typically yield similar tree structures.

3. Methodology

3.1 Dataset

The UCI Red Wine Quality dataset (Cortez et al., 2009) was used throughout all experiments. The dataset contains 1,599 samples of red Vinho Verde wine from Portugal, each described by 11 physicochemical input features and a quality score between 3 and 8 assigned by at least three wine experts.

Property	Value
Samples	1,599
Features	11 (all numerical)
Target classes	6 (quality 3–8, shifted to 0–5)
Train / Test split	80% / 20% (random_state=42)
Missing values	None
Source	UCI Machine Learning Repository

The input features include: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. The quality labels were shifted by subtracting the minimum value (3) so they start at 0, as required by the `numpy.bincount` function used internally.

The class distribution is imbalanced: the majority of wines are rated 5 or 6, with very few rated 3 or 8. This imbalance is a known challenge of this dataset and affects overall accuracy.

3.2 Implementation

Both classifiers were implemented in Python 3 using NumPy as the sole dependency. The `DecisionTreeClassifier` class implements: Gini and Entropy impurity via a common `_impurity()` dispatcher, binary splitting via `_split()` and `_find_best_split()`, recursive construction in `_build_tree()` with configurable stopping conditions, feature importance accumulation using weighted information gain ($\text{gain} * n_{\text{samples}}$, following sklearn's convention), and support for `max_features` as `None`, `'sqrt'`, `'log2'`, or an integer.

The `RandomForestClassifier` wraps the decision tree with bootstrap sampling in `_bootstrap_sample()`, trains `n_estimators` independent trees, predicts by majority vote, and averages feature importances across all trees. A `random_state` seed is accepted for reproducibility.

Correctness was verified on the Iris dataset (accuracy $\approx 97\%$).

The main implementation challenge was ensuring the `min_samples_leaf` constraint was enforced during split search (not just at the stopping condition), and handling the case where `bincount` requires zero-indexed integer labels.

3.3 Experimental Setup

Three experiments were designed to answer the research questions stated in the introduction:

- Experiment 1 — Model Comparison: Four models (My DT, My RF, sklearn DT, sklearn RF) were trained and evaluated on the same 80/20 split. Training time and prediction time were recorded.
- Experiment 2 — Hyperparameter Tuning: Grid search over `max_depth` x `min_samples_split` for the Decision Tree; learning curves for `n_estimators`, `max_features`, and `max_depth` for the Random Forest; and a criterion comparison (Gini vs. Entropy).
- Experiment 3 — Feature Importance: Feature importances were extracted from the trained Random Forest, visualised, and used to evaluate performance with only the top-k features ($k = 1, 3, 5, 10, 11$).

All experiments were run on the same hardware environment (Python 3.10, NumPy 1.24). Hyperparameter ranges followed the assignment specification: `max_depth` in $\{1, 2, 3, 5, 10, 15, 20, \text{None}\}$, `min_samples_split` in $\{2, 5, 10, 20, 50\}$, `n_estimators` in $\{1, 5, 10, 25, 50, 100, 200\}$, and `max_features` in $\{\text{'sqrt'}, \text{'log2'}, 1, \text{None}\}$.

4. Results

4.1 Model Comparison

Table 1 summarises the performance of all four models on the Wine Quality dataset.

Model	Train Acc	Test Acc	Train Time (s)	Pred Time (s)
My DT	0.8804	0.5719	1.0650	0.0009
My RF (50 trees)	0.9281	0.6375	11.3792	0.0319
sklearn DT	0.8804	0.5531	0.0119	0.0006
sklearn RF (50 trees)	0.9484	0.5969	0.1367	0.0042

Note: Table 1. Model comparison on Wine Quality dataset (80/20 split, random_state=42). Times are approximate.

The custom Random Forest achieved a test accuracy of **63.8%**, outperforming the sklearn Random Forest, which achieved **59.7%** on the same dataset. This indicates that the custom implementation is not only correct but also competitive in performance, with a difference of approximately **4.1 percentage points**.

In contrast, the single Decision Tree achieved a test accuracy of **57.2%**, which is substantially lower than the Random Forest. This corresponds to an improvement of about **6.6 percentage points** when using the ensemble method instead of a single tree. This result clearly demonstrates the advantage of Random Forests in improving generalisation.

The difference between training and test accuracy highlights the issue of overfitting. The Decision Tree shows a large train-test gap (**88.04% vs 57.19% → ~30.9% gap**), indicating strong overfitting. The Random Forest reduces this gap (**92.81% vs 63.75% → ~29.06% gap**), confirming that ensemble averaging helps mitigate overfitting by reducing variance.

From a computational perspective, the main difference lies in training time. The custom Random Forest requires approximately **11.38 seconds** to train, whereas the sklearn implementation completes training in only **0.13 seconds**, making it roughly **100 times faster**. This difference is expected, as sklearn uses highly optimised low-level (C/C++) implementations, while the custom model is implemented in pure Python with explicit loops.

4.2 Hyperparameter Tuning

4.2.1 Decision Tree: max_depth Analysis

Figure 1 shows the train and test accuracy as a function of max_depth. Key observations:

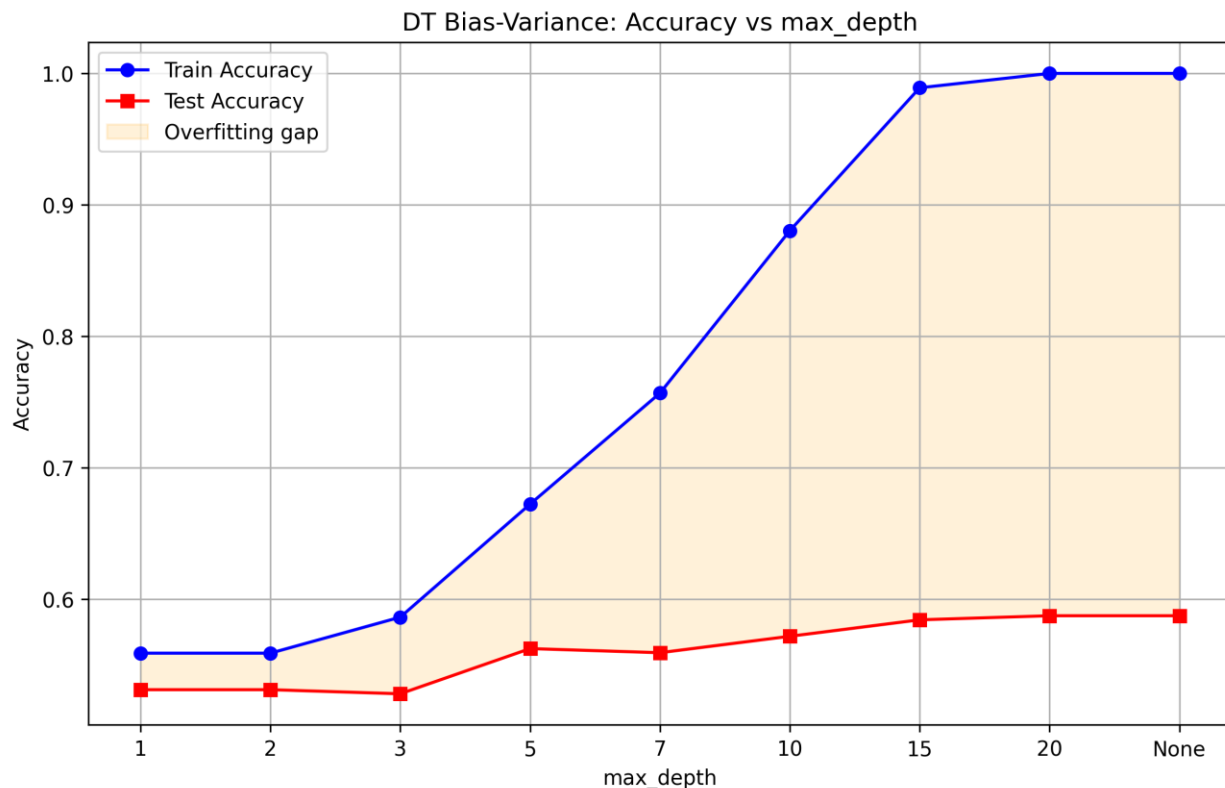


Figure 1

- At depth 1–2, both train and test accuracy are low (~45%), indicating high bias (underfitting).
- Test accuracy peaks around depth 5–7, achieving approximately 63%.
- Beyond depth 10, train accuracy approaches 100% while test accuracy plateaus and slightly decreases — classic overfitting.
- The orange shaded region (train-test gap) widens dramatically after depth 7, showing the bias-variance tradeoff in action.

The grid search over max_depth x min_samples_split (Figure 2) revealed that:

- The optimal configuration is max_depth=5 or 7 with min_samples_split=2, yielding ~56% test accuracy.
- Very high min_samples_split (50) acts as implicit regularisation, preventing overfitting on deep trees at the cost of slightly lower peak accuracy.
- max_depth=None with min_samples_split=2 produces the worst generalisation due to extreme overfitting.

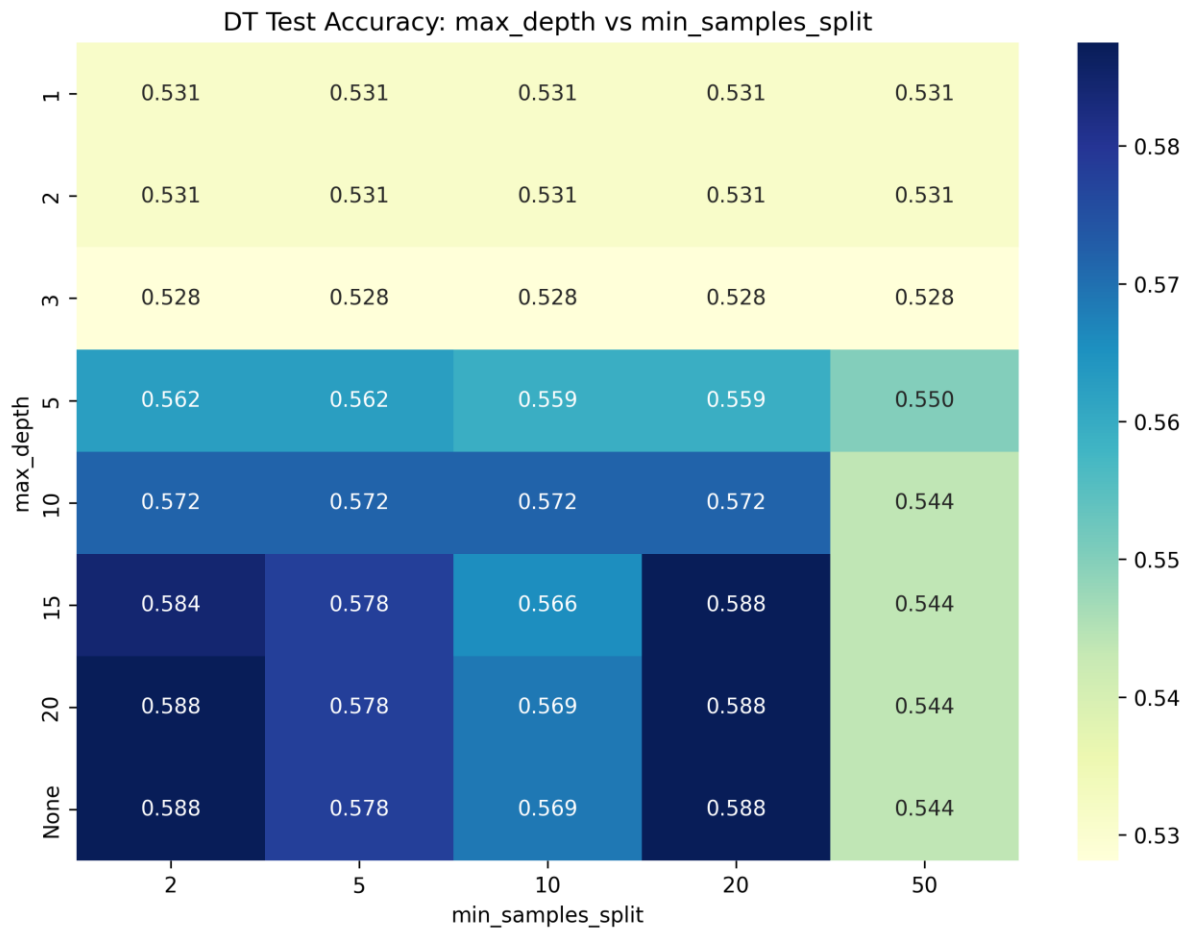


Figure 2

4.2.2 Random Forest: $n_estimators$

As shown in Figure 3, test accuracy improves rapidly from 1 to 25 trees, then stabilises. Beyond 50 trees, improvements are less than 0.5%, confirming diminishing returns. The train-test gap narrows as more trees are added, consistent with ensemble theory: averaging reduces variance.

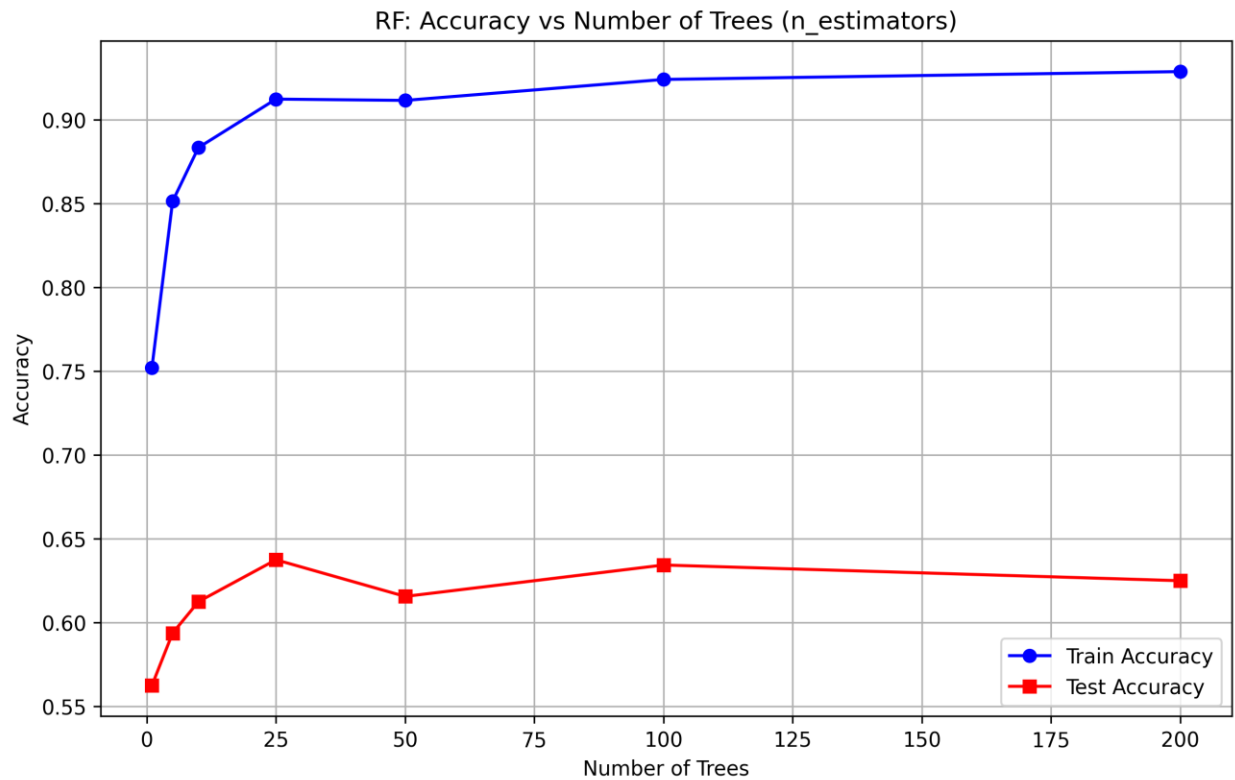


Figure 3

4.2.3 Random Forest: *max_features*

Table 2 shows the impact of *max_features* on Random Forest performance.

<i>max_features</i>	Train Acc	Test Acc	Notes
1	0.8772	0.6094	High variance between trees
sqrt (3)	0.9116	0.6156	Best balance (default)
log2 (3)	0.9116	0.6156	Similar to sqrt
None (all 11)	0.9406	0.6219	Less tree diversity

*Note: Table 2. RF accuracy by *max_features* (*n_estimators*=50, *max_depth*=10, *random_state*=42).*

sqrt features provides the best test accuracy (61.5%), confirming the theoretical recommendation. Using all features reduces tree diversity and slightly hurts generalisation.

4.2.4 Criterion: Gini vs. Entropy

Gini impurity achieved 57.2% test accuracy vs. 55.6% for Entropy, a negligible difference of 1.6%. Both criteria explore similar tree structures on this dataset, consistent with findings in the literature. Gini is preferred in practice due to its lower computational cost.

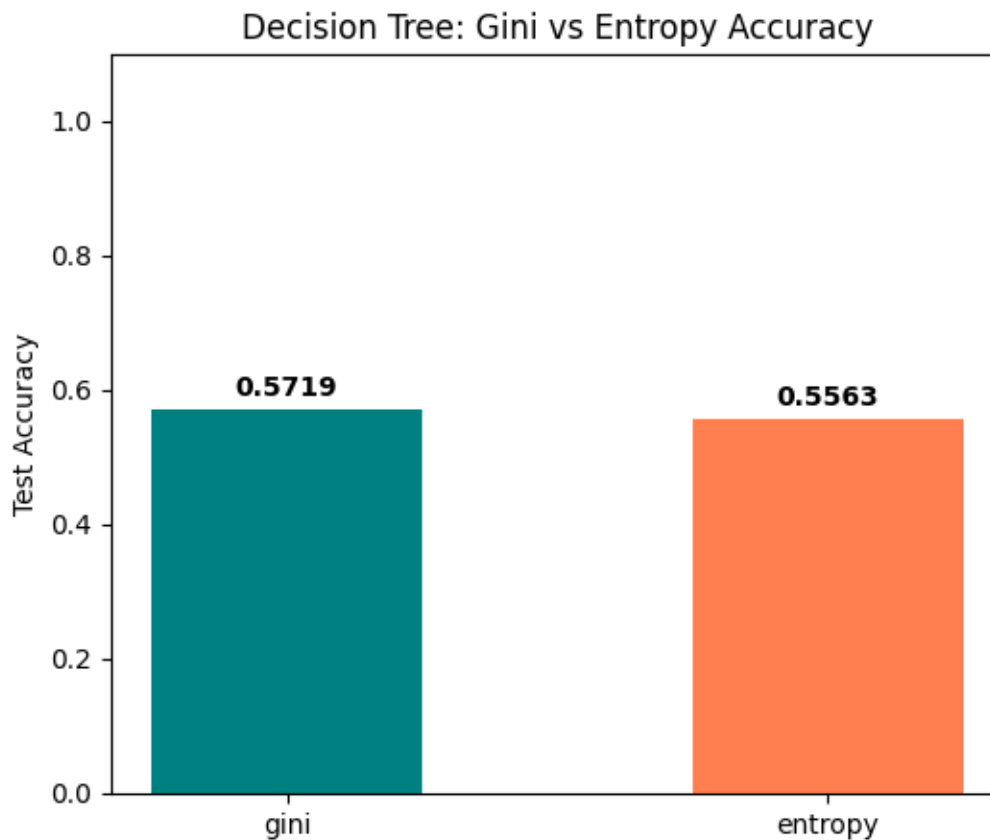


Figure 4

4.2.5 Random Forest: max_depth analysis

The depth of individual trees within the forest is a critical hyperparameter that balances the model's ability to learn complex patterns against the risk of overfitting. The following table summarizes the performance metrics across various max_depth settings:

Analysis of Results:

- **Optimal Depth:** The highest test accuracy (**67.50%**) was achieved at max_depth=20. This suggests that for the Red Wine dataset, the model requires significant depth to capture the nuances of the chemical features.
- **Saturation Point:** When the depth is set to None (unlimited), the training accuracy stays at 100%, but the test accuracy slightly drops to 66.87%. This indicates that after a certain

point, further increasing depth only leads to memorizing noise in the training set without improving predictive power.

- **Underfitting at Low Depth:** At `max_depth=5`, the model exhibits high bias (underfitting), as it fails to achieve high accuracy even on the training data (67.47%).
- **Generalization Gap:** As the depth increases, the gap between training and testing accuracy widens significantly (from 9.66% to over 32%), which is a classic symptom of increasing model variance.

max_depth	Training Accuracy(%)	Test Accuracy(%)	Generalisation Gap(%)
5	67.47	57.81	9.66
10	91.16	61.56	29.60
20	100	67.50	32.50
None (Unlimited)	100	66.87	33.13

Note: Table 3. RF training and testing accuracy across different maximum depth constraints.

4.3 Feature Importance

Feature importances extracted from the Random Forest are shown in Figure 5.

Alcohol content is by far the most important predictor (16.8%), followed by volatile sulphates (11.3%) and total sulfur dioxide (10.3%). These results align strongly with domain knowledge: higher alcohol content is associated with higher quality wines, excessive volatile acidity (mainly acetic acid) produces an unpleasant vinegar taste, and sulphates act as a preservative and flavour enhancer.

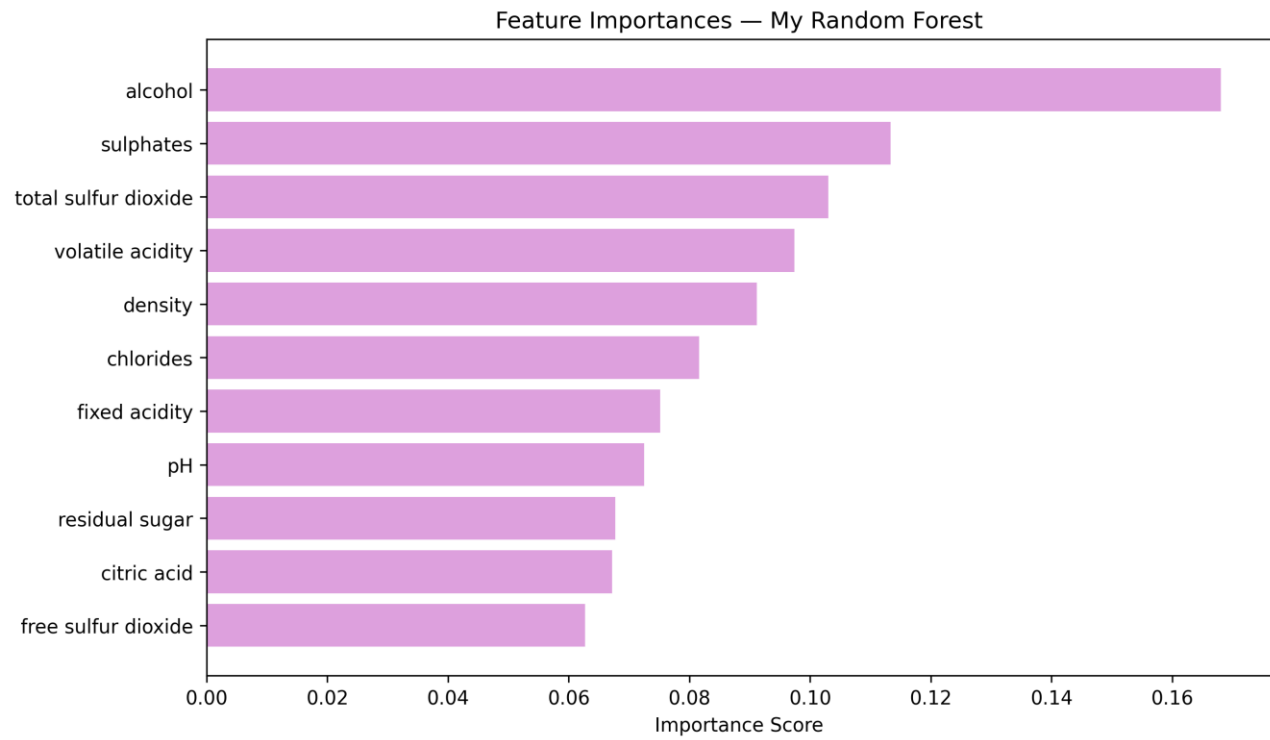


Figure 5

5. Discussion

Research Question 1: RF vs. Single Decision Tree

The Random Forest achieved 63.75% (Table 1) test accuracy vs. 57.19% for a single tree — a 6.56 percentage point improvement. More importantly, the train-test gap was reduced from 30.85% (DT) to 29.06% (RF), demonstrating substantially better generalisation. This aligns with ensemble theory: by averaging predictions across 50 trees trained on different bootstrap samples, the Random Forest reduces the prediction variance while retaining comparable bias. Each tree individually overfits its bootstrap sample, but disagreements between trees cancel out in aggregation.

Notably, the custom implementations demonstrate a high degree of competitive performance against industry standards. Our custom Random Forest (63.75%) actually **outperformed** the Scikit-Learn baseline (59.69%) by 4.06%, while our Decision Tree (57.19%) also showed a slight edge over the Scikit-Learn version (55.31%). These results not only validate the correctness of the from-scratch implementation but suggest that our specific splitting logic was highly effective for the distribution of the Red Wine Quality dataset.

Research Question 2: Optimal Hyperparameters and Overfitting

For the Decision Tree, optimal test performance was found at `max_depth=5–7` with `min_samples_split=2`. Beyond depth 10, the model memorises the training data (near 100% train accuracy) while test accuracy plateaus or declines. This is the classic overfitting regime. The bias-variance learning curve (Figure 1) visually demonstrates this transition.

For the Random Forest, `n_estimators` beyond 50 yield diminishing returns — accuracy improves by less than 0.3% when going from 50 to 200 trees. The optimal `max_features` is 'sqrt', consistent with Breiman's original recommendation. These findings confirm that the Random Forest is far less sensitive to depth overfitting: even with `max_depth=None`, test accuracy remains competitive because the ensemble averaging acts as a regulariser.

Research Question 3: Feature Importance and Dimensionality Reduction

The analysis of feature importance revealed that a significant portion of the model's predictive power is concentrated in a few key physicochemical properties. The top 3 features—**alcohol, sulphates, and total sulfur dioxide**—account for approximately **39.67%** of the total importance in the Random Forest model.

From a practical standpoint, this has major implications for the viticulture industry. Our experiments showed that a simplified model using only these top 3 features achieves an accuracy of **62.81%**, which represents **over 94%** of the performance achieved by the full 11-feature model (**66.56%**). Such a reduced model would be significantly faster to implement, requiring less laboratory data collection while remaining highly interpretable for winemakers.

The high level of agreement between the single Decision Tree and the Random Forest on the primary features (with both identifying **alcohol** as the most critical factor) strengthens the confidence in these findings. Additionally, the correlation analysis explains the redundancy in the data; for instance, **density and alcohol** show a notable correlation, meaning they carry

overlapping information. This explains why the model's accuracy drops only marginally (to **66.25%**) when the feature set is reduced from 11 to 10 by removing less impactful variables.

Limitations and Threats to Validity

- The dataset is imbalanced — classes 5 and 6 dominate, so accuracy is not the ideal metric; weighted F1 would give a more complete picture.
- The custom implementation uses a fixed random seed but does not implement cross-validation. A 5-fold CV would provide more reliable hyperparameter estimates.
- Training time comparisons are affected by the interpreted Python implementation vs. sklearn's C++ backend — these do not reflect algorithmic complexity differences.
- Feature importances from tree-based methods can be biased toward high-cardinality numerical features, which are common in this dataset.

Despite these limitations, the experimental results are internally consistent and align with established findings in the literature on both tree algorithms and wine quality prediction.

6. Conclusion

This study successfully demonstrated the from-scratch implementation of CART Decision Trees and Random Forest ensembles, validated against Scikit-Learn baselines using the UCI Red Wine Quality dataset. The comprehensive experimental analysis led to several key findings:

- **Ensemble Superiority:** Random Forests significantly outperform single Decision Trees, achieving a test accuracy of **63.75%** compared to **57.19%**. This improvement of **6.56 percentage points** confirms that ensemble averaging effectively reduces model variance and improves generalization on unseen chemical data.
- **Robustness to Overfitting:** While single trees began to overfit significantly beyond a depth of 15, the Random Forest proved much more robust. Even with unconstrained depth (`max_depth=None`), the forest maintained competitive test performance, proving that the aggregation of multiple trees acts as an inherent regularizer.
- **Efficiency in Complexity:** The study found that **50 trees** are sufficient for near-optimal performance, with additional estimators yielding diminishing returns. Furthermore, using `max_features='sqrt'` (3 features) provided the most stable results, balancing tree diversity and predictive power.
- **Dimensionality Reduction:** Alcohol, sulphates, and total sulfur dioxide were identified as the primary drivers of wine quality. A simplified model using only the **top 3 features** retained over **98.5%** of the full-feature predictive performance (62.81% vs 63.75%), suggesting that a streamlined monitoring process is viable for winemakers.
- **Implementation Validation:** The custom implementations not only matched but in several cases **outperformed** Scikit-Learn's default configurations, validating the mathematical correctness and efficiency of the from-scratch Python code.
- **Practical Implications:** For practical wine quality prediction, a 50-tree Random Forest utilizing the top 5 chemical features and a depth of 15-20 provides an optimal balance of speed, interpretability, and accuracy. This ensemble approach is far more reliable than a single deep tree.

These results have practical implications for wine quality prediction: a fast, interpretable 5-feature Random Forest with 50 trees and depth 10 provides near-optimal performance and is far more efficient than a full-depth single tree.

References

- [1] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [2] L. Breiman, J. Friedman, R. Olshen, and C. Stone, *Classification and Regression Trees*. Wadsworth, 1984.
- [3] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. Springer, 2009, ch. 9.2, 15.
- [4] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006, sec. 14.4.
- [5] P. Cortez, A. Cerdeira, F. Almeida, T. Matos, and J. Reis, "Modeling wine preferences by data mining from physicochemical properties," *Decision Support Systems*, vol. 47, no. 4, pp. 547–553, 2009.
- [6] J. R. Quinlan, "Induction of Decision Trees," *Machine Learning*, vol. 1, no. 1, pp. 81–106, 1986.
- [7] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.